

Expanded Polystyrene Exposure and Nitrite-Related Absorbance in *Eisenia fetida*

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ABSTRACT

Microplastic contamination can affect soil properties, microbial communities, and nutrient cycling. This study evaluated whether expanded polystyrene (EPS) microplastic exposure during vermicomposting affected nitrite-related absorbance in vermicast produced by *Eisenia fetida*. Fifty independent experimental units were divided into a control treatment without EPS (n = 25) and an EPS treatment (n = 25). After a 12-day exposure period, vermicast samples were extracted with calcium chloride and analyzed using a modified Griess-type colorimetric reaction. Griess-reactive absorbance was measured at 540.1 nm with a visible-light spectrophotometer. Mean absorbance was 1.25 absorbance units (AU) in the EPS treatment and 0.766 AU in the control, a 63% difference between group means. A Mann-Whitney U test indicated a significant difference between treatments (U = 75, z = -4.6, p < 0.01). Under these experimental conditions, EPS exposure was associated with greater Griess-reactive absorbance in vermicast. Because no nitrite calibration curve was generated, the results cannot be interpreted as absolute nitrite concentrations, and the mechanism underlying the observed difference remains unresolved.

Keywords: expanded polystyrene; *Eisenia fetida*; microplastics; vermicomposting; nitrite-related absorbance; Griess assay

INTRODUCTION

Soil is a fundamental component of terrestrial ecosystems, supporting plant growth, regulating water and nutrient cycling, and sustaining diverse biological communities. However, increasing anthropogenic activities have led to widespread soil pollution, posing a growing threat to soil health and ecosystem functioning. Among soil organisms, *Eisenia fetida* (commonly known as red wigglers or tiger worms) is an epigeic earthworm widely used in studies of soil ecology and vermicomposting because of its essential role as an ecosystem engineer. Through burrowing, casting, and the decomposition of organic matter, *E. fetida* improves soil aeration, enhances water retention, stimulates microbial activity, and promotes nutrient cycling (Ya et al. 2021). In particular, earthworm-mediated decomposition accelerates the mineralization of organic nitrogen into plant-available forms, thereby improving soil fertility and supporting plant productivity (Wang et al. 2024). These ecological functions underpin the widespread application of vermicomposting, in which *E. fetida* contributes to the processing of organic wastes (Cui et al. 2022).

Despite the well-established ecological benefits of *E. fetida*, the increasing accumulation of plastic debris in terrestrial ecosystems presents a growing challenge to soil ecosystems. Expanded polystyrene (EPS), commonly known as Styrofoam, is one of the most prevalent plastic materials found in the environment because of its extensive use in packaging, disposable food containers, insulation, and other consumer products. Although EPS is technically recyclable, low recycling rates have resulted in its widespread environmental accumulation (Qin et al. 2021). As larger plastic materials weather and fragment, they generate microplastics (MPs), which are generally defined as plastic particles smaller than 5 mm in diameter. MPs are now ubiquitous in agricultural soils, urban environments, and industrial regions worldwide (Rillig et al. 2017; Qin et al. 2021). Owing to their persistence and small size, MPs can alter soil physical properties, reduce water infiltration, modify microbial community composition, and influence nutrient availability, thereby posing substantial risks to soil ecosystem functioning (Qin et al.

2021).

Because earthworms continuously ingest soil and organic matter, they are inevitably exposed to MPs present in contaminated soils. Previous studies have detected microplastics within earthworm digestive systems and casts, indicating that *E. fetida* can ingest and redistribute plastic particles during the processing of soil and organic matter (Rillig et al. 2017; Cui et al. 2022). Mechanical processing within the earthworm digestive tract may reduce the size of ingested particles without completely degrading the polymer, thereby contributing to the formation of smaller secondary microplastics (Ya et al. 2021; Cui et al. 2022). However, the extent to which *E. fetida* specifically alters EPS particle size remains insufficiently characterized.

Microplastics may also influence soil nitrogen transformation by altering soil physicochemical properties, nutrient availability, and microbial communities involved in processes such as nitrification and denitrification (Wang et al. 2024). These effects may occur through several pathways, including changes in microbial habitat and activity as well as interactions between plastic surfaces and chemical species. Consequently, the influence of EPS microplastic exposure on nitrogen-related chemical properties of vermicast remains poorly understood. This study therefore evaluated whether exposure to expanded polystyrene microplastics during vermicomposting affected nitrite-related absorbance in vermicast produced by *E. fetida*. It was hypothesized that vermicast from EPS-treated experimental units would exhibit greater Griess-reactive absorbance at 540.1 nm than vermicast from untreated control units.

MATERIALS AND METHODS

Experimental Design and Vermicomposting Conditions

A total of 50 square plastic pots were used as independent experimental units, with 25 assigned to the control treatment and 25 assigned to the expanded polystyrene (EPS) treatment. Each pot was labeled according to treatment and replicate number, and the drainage openings at

the bottom of the pots were sealed with duct tape. Bedding was prepared from shredded egg-carton material moistened with reverse-osmosis water. Approximately 50 mL of reverse-osmosis water was added at a time and mixed into the bedding until the material was uniformly damp without visible excess water. Prepared bedding was added to each pot to approximately half of the pot depth.

A total of 12.5 g of *Eisenia fetida* was weighed using an analytical balance and introduced into each experimental unit. After the earthworms had settled, they were covered with approximately 5 cm of additional bedding. Perforated cardboard covers were placed over the pots, and the earthworms were allowed to acclimate and burrow into the bedding. All 50 experimental units were placed on flats without drainage holes and maintained in the same classroom environment. Nine days after introduction of *E. fetida*, one portion of frozen watermelon rind was added to each experimental unit as a food source. The units were monitored every other school day when access to the classroom was available. Additional watermelon rind was provided after the previously supplied food had been consumed.

EPS Preparation and Exposure

Expanded polystyrene (EPS) obtained from discarded packaging foam was mechanically fragmented before experimental exposure. A total of 5 g of EPS was prepared, and 0.2 g portions were weighed using an analytical balance for each of the 25 EPS-treatment units. The particles were prepared to fall within the operational microplastic size range of less than 5 mm; however, a quantitative particle-size distribution was not determined. Eighteen days after introduction of *E. fetida*, 0.2 g of prepared EPS was added to each of the 25 EPS-treatment units. The 25 control units received no EPS. The contents of the experimental units were subsequently mixed.

The experimental units were maintained for an additional 12 days. During this period, the units were maintained weekly by mixing and by adding reverse-osmosis water, food, and shredded egg-carton material as required to maintain suitable vermicomposting conditions.

Substrate temperature was monitored periodically using digital soil thermometers; however, quantitative temperature records were not retained. After the 12-day exposure period, feeding was discontinued, and 2.5 g of vermicast was collected from each experimental unit for chemical analysis.

Vermicast Extraction

A calcium chloride extraction solution was prepared by dissolving 1.84 g of calcium chloride dihydrate in reverse-osmosis water to a final volume of 500 mL. Each 2.5 g vermicast sample was transferred to a 150 mL beaker labeled with the date, sample identity, treatment designation, and researcher initials (J.P. or L.G.). Calcium chloride extraction solution (10 mL) was then added to each beaker. Each sample was manually mixed with a glass stirring rod for 1 min. Activated charcoal (0.5 g) was subsequently added to each sample. The beakers were covered with aluminum foil and mixed on a mixing hot plate at 450 rpm for 30 min. Following mixing, the samples were allowed to settle for 10 min and were then filtered through folded filter paper into labeled 50 mL beakers.

Griess Reagent Preparation and Color Development

Nitrite-related absorbance was evaluated using a modified Griess-type colorimetric reaction based on diazotization with sulfanilamide and coupling with N-(1-naphthyl)ethylenediamine (NED). This reaction produces an azo chromophore with strong absorbance near 540 nm, consistent with established colorimetric methods for nitrite determination (U.S. Environmental Protection Agency 1993). The color reagent used in this study was prepared by combining 177.78 mL of reverse-osmosis water with 22.22 mL of phosphoric acid solution under a chemical fume hood. Sulfanilamide (8.89 g) was added and mixed until dissolved, followed by 0.22 g of N-(1-naphthyl)ethylenediamine (NED) reagent. The reagent was allowed to stand for 30 min before use. A 3 mL aliquot of the prepared color reagent was added to each filtered vermicast extract, and the reaction mixtures were allowed to develop for 10 min before spectrophotometric analysis.

Reagent Blank

A reagent blank was prepared using 10 mL of calcium chloride extraction solution and 0.5 g of activated charcoal and was processed using the same mixing, settling, filtration, and color-development procedure as the vermicast samples, except that no vermicast was included. Spectrophotometric Analysis

Following color development, each sample was transferred to a correspondingly labeled cuvette for spectrophotometric measurement. A Vernier visible-light (VIS) spectrophotometer was used to measure the absorbance of each prepared sample. The reagent blank was used to zero the spectrophotometer before sample measurements. Absorbance was measured and recorded at 540.1 nm for each experimental unit.

No external nitrite standards or multi-point calibration curve were used during the original analysis. Therefore, the measurements were evaluated as relative Griess-reactive absorbance rather than converted to absolute nitrite concentrations.

Statistical Analysis

Each pot was treated as one independent experimental unit, resulting in 25 replicates for the control treatment and 25 replicates for the EPS treatment. Descriptive statistics, including the mean and standard deviation of absorbance values, were calculated for each treatment group. Differences in Griess-reactive absorbance between the control and EPS-treatment groups were evaluated using a Mann-Whitney U test in DataClassroom. The resulting U statistic, z statistic, and p-value were reported.

RESULTS

Vermicast from EPS-treated experimental units exhibited greater Griess-reactive absorbance at 540.1 nm than vermicast from untreated units. Mean absorbance was 1.25 absorbance units (AU) in the EPS treatment group and 0.766 AU in the control group, corresponding to a 63% difference between group means. A Mann-Whitney U test indicated a

statistically significant difference between the treatment groups ($U = 75$, $z = -4.6$, $p < 0.01$). The standard deviation was 0.3512 AU for the EPS treatment group and 0.249 AU for the control group. Individual experimental-unit values and group summary statistics are shown in Figure 1. Descriptive statistics for both treatment groups are summarized in Table 1, and the results of the Mann-Whitney U test are presented in Table 2.

DISCUSSION

This study examined whether the presence of expanded polystyrene (EPS) microplastics during vermicomposting affected nitrite-related absorbance in vermicast produced by *Eisenia fetida*. The EPS treatment group had a mean absorbance of 1.25 AU at 540.1 nm, compared with 0.766 AU in the control group, representing a 63% increase. The Mann-Whitney U test indicated a statistically significant difference between the groups ($U = 75$, $z = -4.6$, $p < 0.01$). These findings indicate that exposure to EPS microplastics was associated with greater Griess-reactive absorbance in the resulting vermicast under the conditions of this experiment.

The greater absorbance observed in the EPS treatment group is consistent with previous evidence that microplastics can influence soil physicochemical conditions, microbial communities, and nitrogen transformation processes (Ya et al. 2021; Wang et al. 2024). Microplastics may affect nitrogen cycling through changes in microbial habitat and activity, including processes associated with nitrification and denitrification. However, the present study did not measure microbial community composition or individual nitrogen-transformation pathways; therefore, the mechanism responsible for the observed difference in Griess-reactive absorbance cannot be determined from these data alone.

Several mechanisms could account for the greater Griess-reactive absorbance observed in the EPS treatment group. One biologically plausible explanation is that EPS exposure altered microbial processes involved in nitrification and denitrification, thereby changing the balance between nitrite production and oxidation. Previous experimental research has demonstrated that

microplastic exposure can modify microbial groups associated with nitrogen-cycling processes, including nitrate reduction and nitrite respiration (Wang et al. 2024). A second possibility is that interactions between microplastic surfaces and dissolved nitrogen species altered their availability within the vermicomposting matrix. Alternatively, part of the measured difference may have resulted from analytical or matrix effects rather than a biological change in nitrite. Because microbial communities, individual nitrogen-transformation pathways, EPS leachates, and potential optical interferences were not directly measured, these mechanisms cannot be distinguished by the present experiment. A microbially mediated alteration of nitrogen transformation is consistent with previous soil studies, although analytical interference remains a plausible alternative explanation.

Direct quantitative comparison of the 63% difference observed in this study with previous research is limited because published studies have used different microplastic polymers, concentrations, exposure conditions, matrices, and nitrogen-related endpoints. Zhang et al. (2023) found that polyethylene and polylactic acid microplastics altered microbial functions associated with soil nitrogen cycling, supporting the broader direction of a microplastic-associated change in nitrogen processing, although their endpoints were not directly comparable with Griess-reactive absorbance in vermicast. Chen et al. (2020) demonstrated biological responses of *E. fetida* to low-density polyethylene microplastics but did not evaluate a directly comparable nitrite-related endpoint. Zhou et al. (2024) examined the effects of excessive nitrite accumulation on soil microorganisms under a different experimental context rather than microplastic exposure. Therefore, the 63% difference observed in the present study should be interpreted as specific to this experimental system rather than as an effect size directly comparable with those previous studies.

The original experimental rationale proposed that earthworm-mediated fragmentation of EPS could increase microplastic particle abundance and surface area and thereby influence

nitrogen-related chemistry. However, EPS fragmentation, particle abundance, and particle surface area were not directly quantified in this study. Furthermore, the EPS was mechanically fragmented before its addition to the experimental units. Consequently, the observed difference in absorbance cannot be attributed specifically to fragmentation by *E. fetida*. Detailed characterization of EPS particle size before and after vermicomposting would be required to test this mechanism directly. The experimental design also did not include an abiotic EPS control without earthworms. Consequently, EPS-associated effects occurring independently of *E. fetida*, including interactions between the polymer and the vermicomposting matrix, cannot be separated from worm-mediated processes. In addition, only a single EPS dose of 0.2 g per experimental unit was evaluated, preventing assessment of a dose-response relationship. Treatment identity was known during spectrophotometric analysis because samples were labeled according to treatment group; therefore, absorbance measurements were not performed under blinded conditions. Future studies should therefore include multiple EPS concentrations as well as appropriate abiotic and biological controls.

Several analytical limitations should also be considered. A multi-point nitrite calibration curve was not generated; therefore, the measured absorbance values cannot be converted to absolute nitrite concentrations, and limits of detection and quantification were not determined. The results should consequently be interpreted as differences in relative Griess-reactive absorbance rather than confirmed differences in nitrite concentration. Extract pH was also not measured. Because the Griess colorimetric reaction is sensitive to reaction conditions, potential differences in extract pH or other matrix components between treatments may have contributed to the measured absorbance and cannot be excluded. Potential interference from the sample matrix was also not directly evaluated. EPS-derived compounds, residual turbidity, or fine activated-charcoal particles remaining after filtration could potentially have contributed to measured absorbance independently of the nitrite-derived color reaction. Future studies should

therefore incorporate matrix blanks, appropriate interference controls, and recovery testing with known nitrite standards to distinguish true nitrite-related absorbance from background or matrix-associated optical effects.

Additional limitations include the use of a single EPS source and treatment concentration, the absence of quantitative characterization of the EPS particle-size distribution, and the relatively short 12-day exposure period. Future research should directly characterize EPS particle size and abundance before and after vermicomposting, evaluate multiple environmentally relevant exposure concentrations, measure extract pH, and incorporate nitrite standards for quantitative calibration. Measurements of nitrate, ammonium, microbial community composition, and relevant nitrogen-transformation processes would further clarify whether and how EPS exposure influences nitrogen cycling during vermicomposting.

Under the conditions of this study, vermicast from EPS-treated *Eisenia fetida* experimental units exhibited significantly greater Griess-reactive absorbance at 540.1 nm than vermicast from untreated control units. Mean absorbance was 63% greater in the EPS treatment group, with a statistically significant difference between treatment distributions. Because no nitrite calibration curve was generated, the results cannot be interpreted as absolute nitrite concentrations, and the mechanism responsible for the observed difference remains unresolved. Nevertheless, the findings indicate that EPS exposure was associated with measurable changes in nitrite-related absorbance during vermicomposting and support further investigation using quantitative nitrite calibration, additional experimental controls, direct characterization of microplastic particles, and measurements of nitrogen-cycle processes.

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Table 1. Descriptive statistics for Griess-reactive absorbance (AU) at 540.1 nm in vermicast from control and EPS-treated experimental units.

Treatment	n	Mean	SD	SEM	95% CI	Median	Min-Max
Control (no EPS)	25	0.766	0.249	0.0498	0.663-0.868	0.743	0.433-1.46
EPS	25	1.25	0.3512	0.0703	1.105-1.395	1.175	0.556-1.975

All values are in absorbance units (AU). CI, confidence interval; SD, standard deviation; SEM, standard error of the mean.

Table 2. Mann-Whitney U test comparing Griess-reactive absorbance between control and EPS-treated experimental units.

Comparison	U	z	p	Interpretation
Control vs. EPS	75	-4.6	< 0.01	Griess-reactive absorbance differed significantly between the control and EPS treatment groups.

FIGURE CAPTIONS

Figure 1. Griess-reactive absorbance at 540.1 nm in vermicast from control and EPS-treated experimental units. Individual points represent independent experimental units; the central marker represents the group mean, and error bars represent one standard deviation.

Figure 1

